

St. Andrews Scots Sr. Sec. School

9th Avenue, I.P. Extension, Patparganj, Delhi – 110092.

Class: IX

Ch-9: Force & Laws Of Motion

Topic: Notes

FORCE

Force may be defined as an agency (a push or pull) which changes or tends to change the state of rest or of uniform motion or the direction of motion of a body.

EFFECTS OF FORCE

A force applied on an object can produce the following changes:

1. A force can change a state of rest.

For Example – If we kick a football kept on the ground, then the football starts moving; If we push a resting table, then the table will start moving. In both the cases, the force applied has changed the state of rest.

2. A force can stop a moving body.

For Example – A football moving on ground stops after sometime, because of the force of friction of ground.

3. A force can change the speed of an object.

For Example – A speed of a moving car can be increased by accelerating the car or can be decreased or stopped by applying the brakes.

4. A force can change the direction of motion of an object.

For Example – A force exerted on the steering wheel of a car changes the direction of motion; A force exerted by the cricket player's bat changes the direction of moving cricket ball.

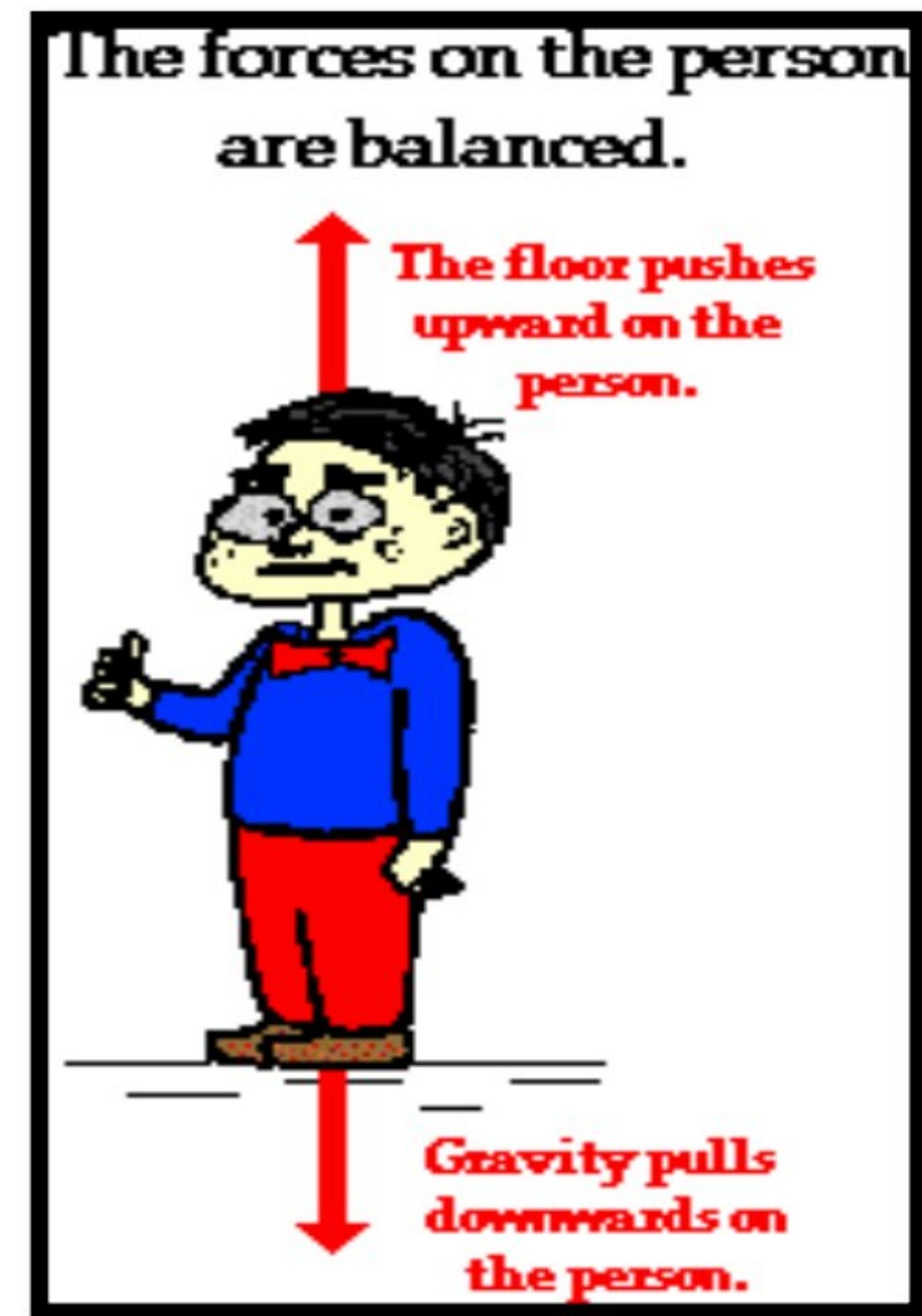
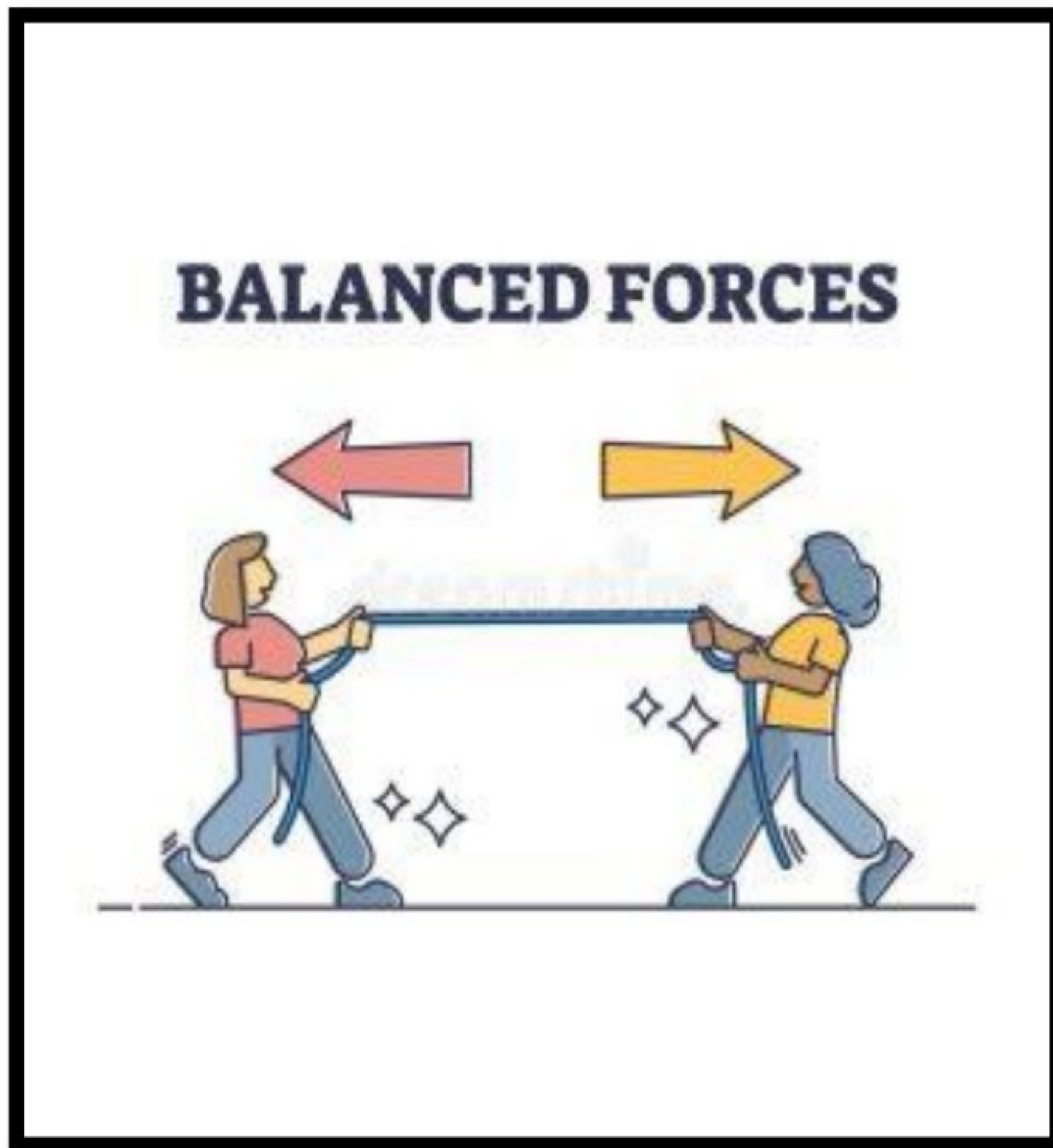
5. A force can change the shape of an object.

For Example – If we pull a spring from both the ends, then the shape of spring changes.

BALANCED FORCES

If the resultant of all the forces acting on a body is zero, the forces are called **balanced forces**.

For Example - A game of tug of war with equally matched opponents is an example of balanced force; When the man pushes against the wall, the forces at play balance each other, and as a result, the wall does not move.



UNBALANCED FORCES

If the resultant of all the forces acting on a body is not zero, the forces are called **unbalanced forces**.

For Example - The ball moves from one place to another after it's kicked.



INERTIA

The inherent property or tendency of an object by virtue of which it cannot change its state of rest or uniform motion in a straight line is called **inertia**.

TYPES OF INERTIA

a) Inertia of rest – The tendency of a body to remain in its state of rest is called inertia of rest.

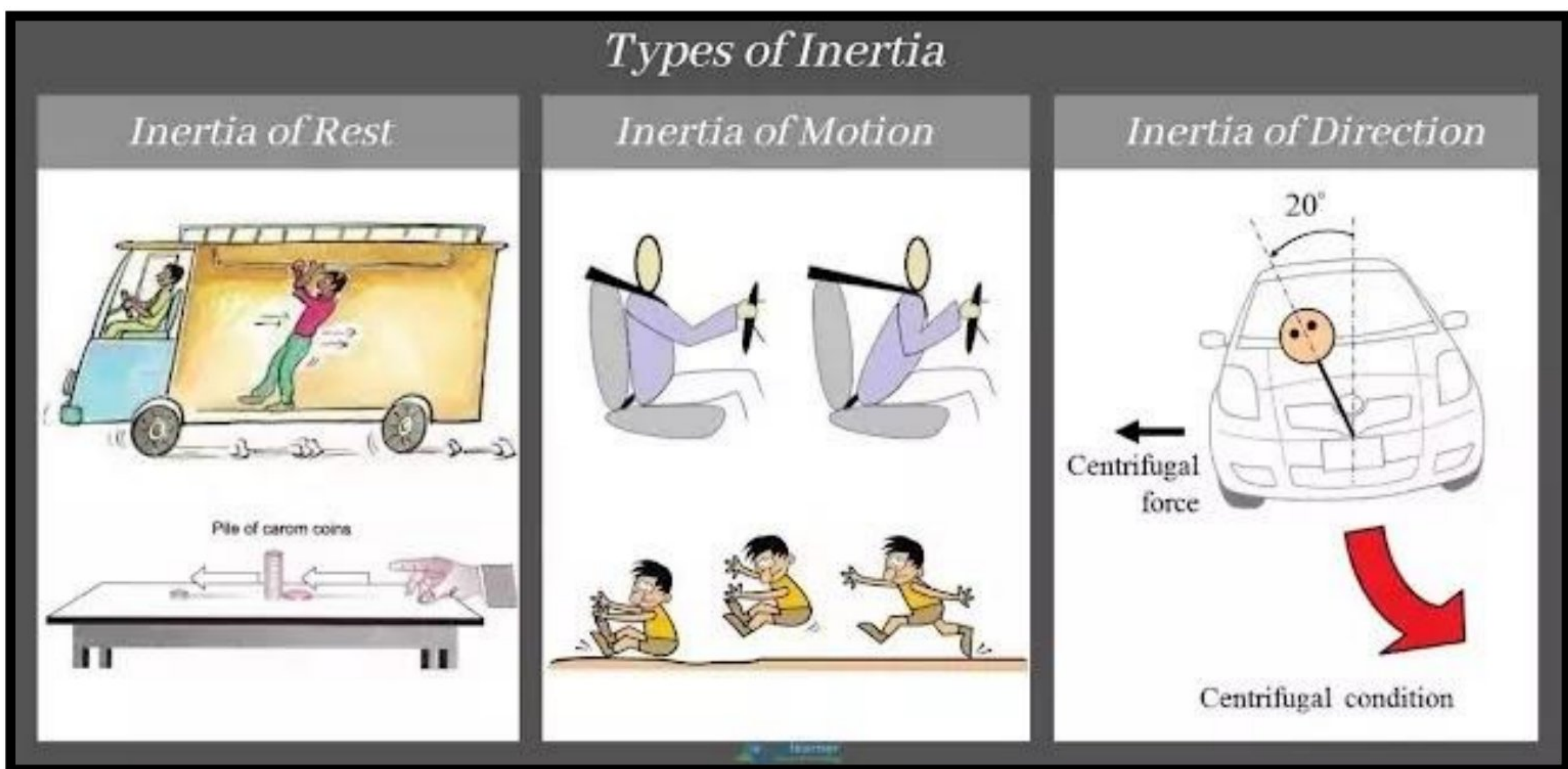
Example – A person standing in a bus falls backward when the bus suddenly starts moving forward. When the bus moves, the lower part of his body begins to move along with the bus while the upper part of his body continues to remain at rest due to inertia. That is why, a person falls backward when the bus starts.

b) Inertia of motion – The tendency of a body to remain in its state of uniform motion in a straight line is called inertia of motion.

Example – When a moving bus suddenly stops, a person sitting in it falls forward. As the bus stops, the lower part of the body comes to rest along with the bus while the upper part of body continues to remain in motion due to inertia and falls forward.

c) Inertia of direction – The inability of a body to change its direction of motion is called inertia of direction.

Example – When a bus takes a sharp turn, a person sitting in the bus experiences a force acting away from the centre of the curved path due to his tendency to move in the original direction.



❖ **Mass as the measure of inertia** – Mass is the measure of its inertia. If a body has more mass, it has more inertia. And if a body has less mass, it has less inertia.

Example – A stone has more inertia than a football; A truck has more inertia than a bicycle; etc.

NEWTON'S LAWS OF MOTION

Sir Isaac Newton (1642-1727) made a systematic study of motion and gave three laws which describe the motion of bodies. These three laws are known as Newton's Laws of Motion.

1. Newton's First Law Of Motion

According to Newton's first law of motion, a body at rest will remain at rest, and a body in motion will continue its uniform motion in a straight line until and unless some external force is applied to change its state of rest or of uniform motion. This law consists of two parts: